Panacea:
A foundation model for clinical trial search, summarization, design, and recruitment

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Motivation

❑ Challenges in Clinical Trials

⚙️ Complex Design Pipeline

- multiple interdependent components
  (trial descriptions, eligibility criteria, study arms, outcome metrics, …)

🔒 Privacy Constraints

- often not amenable to public cloud-based tools (e.g. GPT-4)

👨‍⚕️ Need for Domain Expertise

- Eligibility criteria design & patient recruitment
- Requires specialized knowledge

workflow

❑ Limitations of Existing Methods

📌 Task-specific models

- Designed for individual tasks (e.g., matching, design)

🔗 Fragmented workflow

- Difficult to manage multiple models
- Inefficient use of training data

Core Idea: From General Models to Panacea

💡 **Insight**

General foundation models show strong multi-task capability (e.g. GPT-4 in biomedical tasks)

👉 *Then why not build a specialized one?*

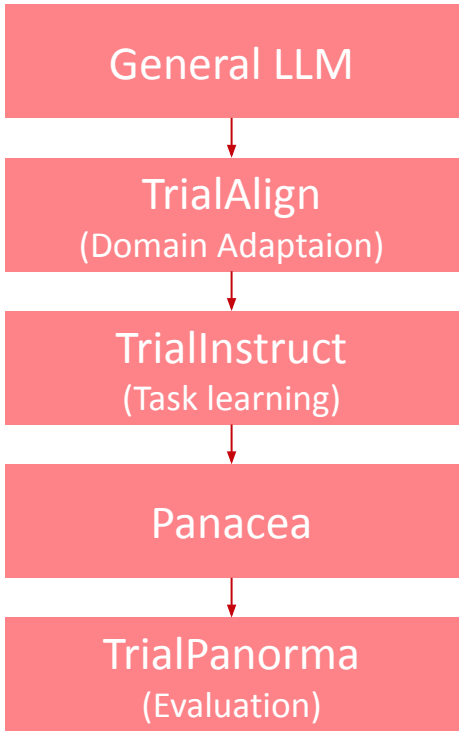
🚀 **Hypothesis**

A specialized foundation model can serve as a **"Swiss Army Knife"** for clinical trial tasks

👉 *This motivates Panacea*

🧠 **Panacea Overview**

- **TrialAlign**: inject clinical knowledge
- **TrialInstruct**: enable multi-task capability
- **TrialPanorama**: comprehensive benchmark



TrialAlign: Domain Knowledge Injection

🧾 **Large-scale Data Collection**

- 793K clinical trial documents from 14 diverse sources
- 1.1M trial-related papers (PubMed, Embase)

👉 Rich clinical + scientific knowledge

🧹 **Data Processing (Quality Control)**

- De-duplication (across sources)
- Remove Personal identifiable information (PII)
- Train/test split (before/after 2023)

👉 Avoid data leakage & ensure data quality

🗂️ **Structured Representation**

- Study overview / Eligibility criteria / Study design / Outcome measures

👉 Mirrors real clinical trial workflow

Used for large-scale alignment training (1.8M samples)

TrialInstruct: Task Learning

🔎 **Trial Search**

- Query generation & Expansion

👉 Generate and refine clinical trial queries using standardized medical terms (MeSH)

📝 **Trial Summarization**

- Single-trial & Multi-trial (Cochrane-based)

👉 Summarize trial documents or multiple studies into concise insights

🧪 **Trial Design**

- Criteria / Arms / Outcomes

👉 Assist trial design through interactive, reasoning-based conversations

🧑‍⚕️ **Patient–Trial Matching**

- Instruction-based Eligibility Prediction

👉 Match patient notes to trial criteria using clinical benchmarks (TREC/SIGIR)

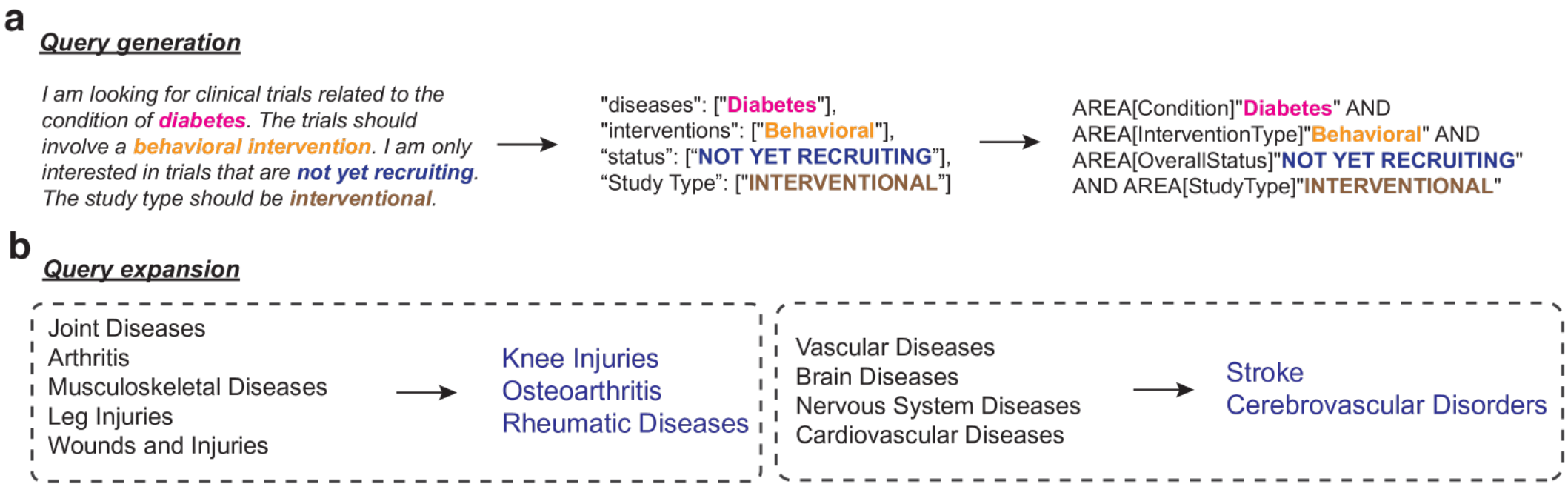
Fine-tuned to enable multi-task capability

- MeSH: Medical Subject Headings
- TREC: Text Retrieval Conference
- SIGIR: Special Interest Group on Information Retrieval

TrialPanorama: Comprehensive Benchmark

Task type	Task name	Metric	Description	Data size		
				Train	Dev	Test
Trial search	Query generation	Jaccard index	Generate searchable queries based on specific clinical trial requirements for database retrieval.	1,837	324	925
	Query expansion	Jaccard index	Broaden search parameters to include related terms and conditions to enhance trial discovery.	43,350	7,650	2,500
Trial summarization	Single-trial summarization	ROUGE, LLM-based metric	Summarize key details and results of individual clinical trials.	4,250	7,50	1,000
	Multi-trial summarization	ROUGE, LLM-based metric	Compile and compare outcomes across multiple clinical trials for comprehensive insights.	1,725	304	252
Trial design	Criteria design	BLEU ROUGE Clinical relevance	Define eligibility criteria for patient selection in clinical trials.	30,559	5,392	549
	Study arm design		Develop different intervention groups to assess the effects of treatments.	45516	8032	549
	Outcome measure design		Establish methods for measuring trial results and effectiveness of interventions.	38,088	6,721	549
Patient-trial matching	Patient-trial matching	F1, BACC, KAPPA	Match eligible patients with suitable clinical trials, 3-class classification problem	24,146	4,261	11,341

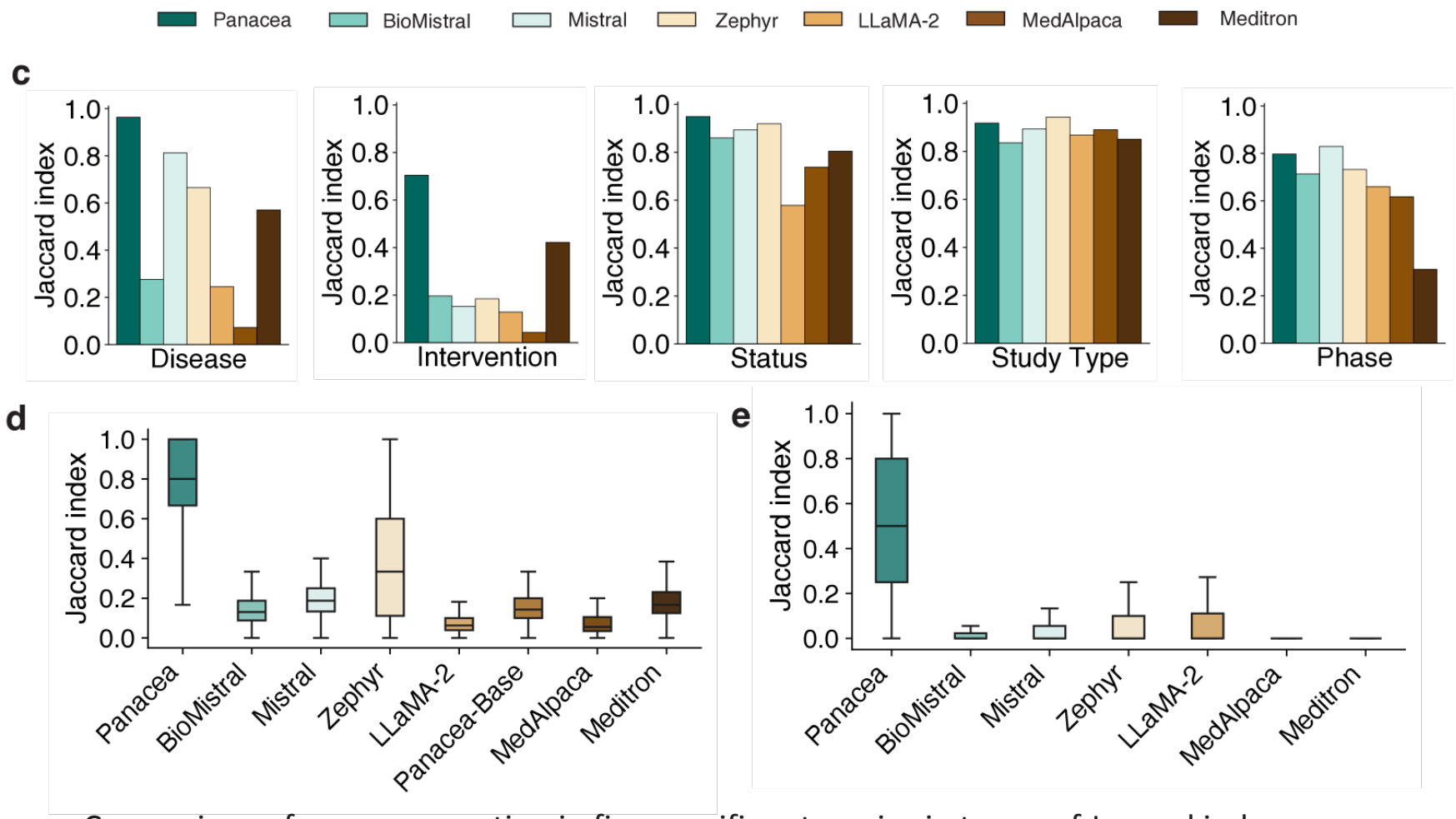
Results: Trial Search



a. Query generation aims to convert free text user input into a structured query that contains five categories: disease, intervention, phase, status, and study type.

b. Query expansion aims to expand a set of keywords. Candidate keywords are not provided.

Results: Trial Search

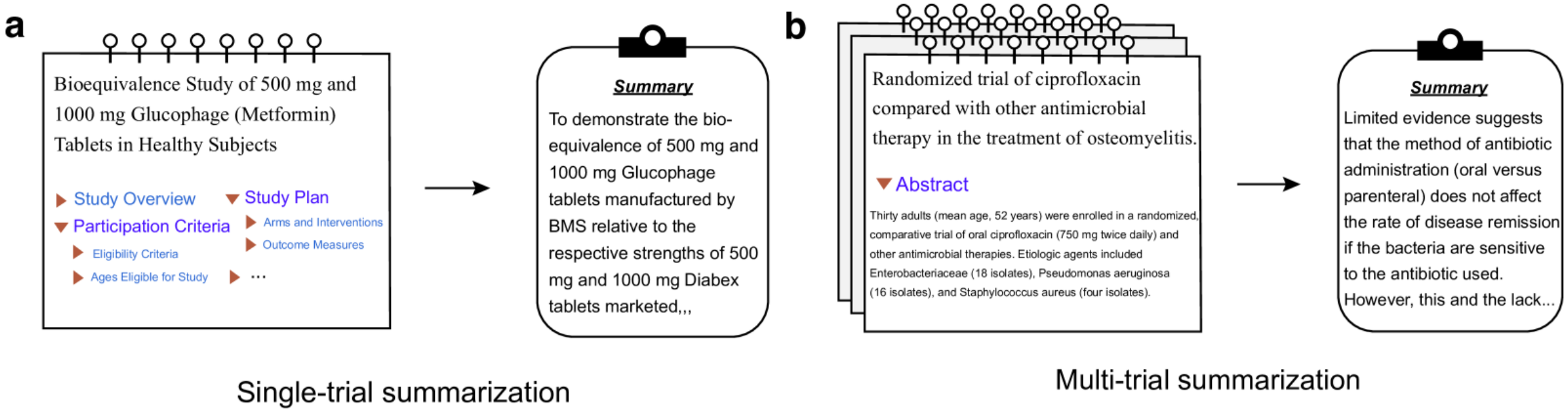


c. Comparison of query generation in five specific categories in terms of Jaccard index.

d. Comparison of query generation in terms of Jaccard index.

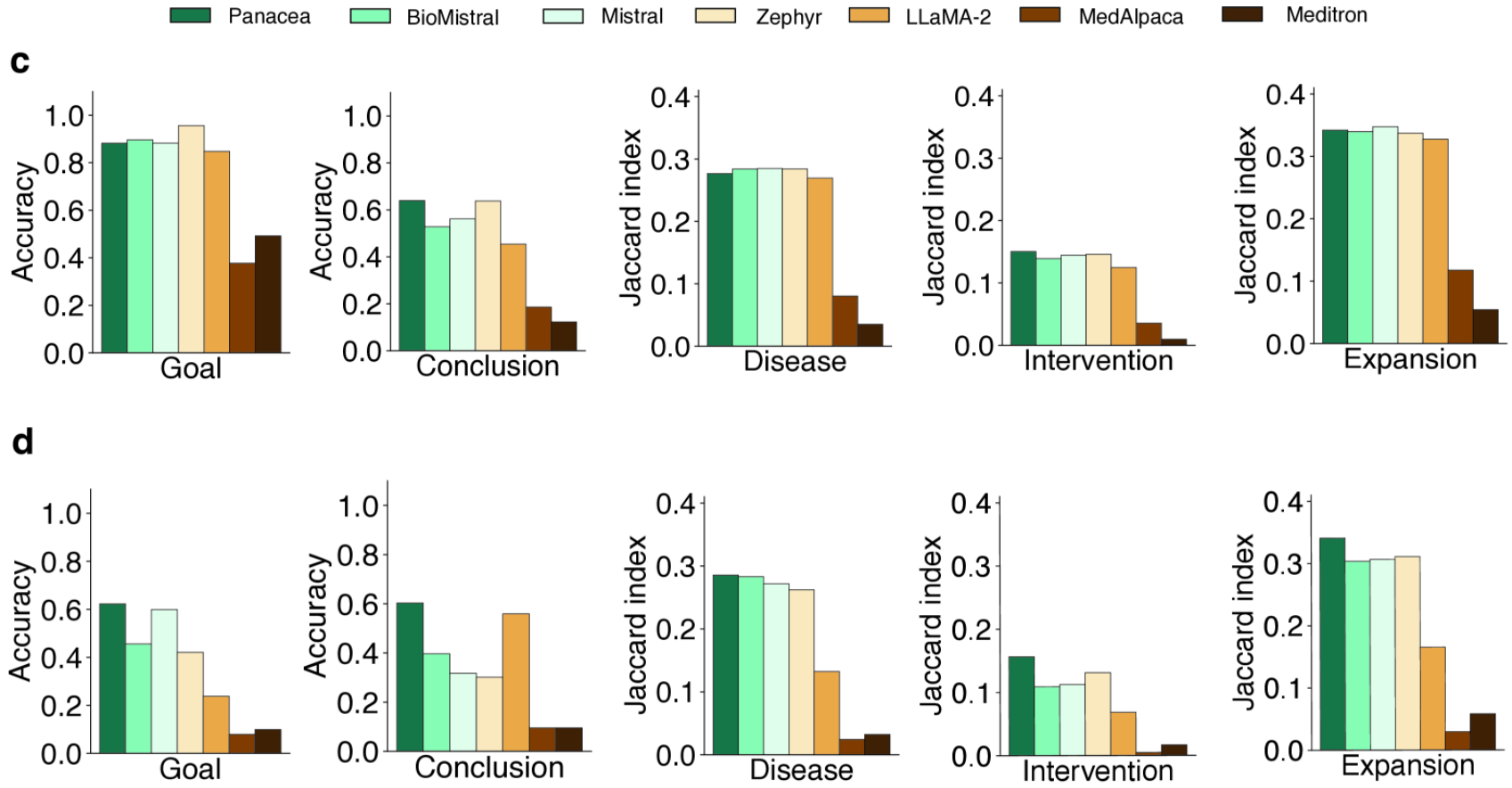
e. Comparison of query expansion in terms of Jaccard index.

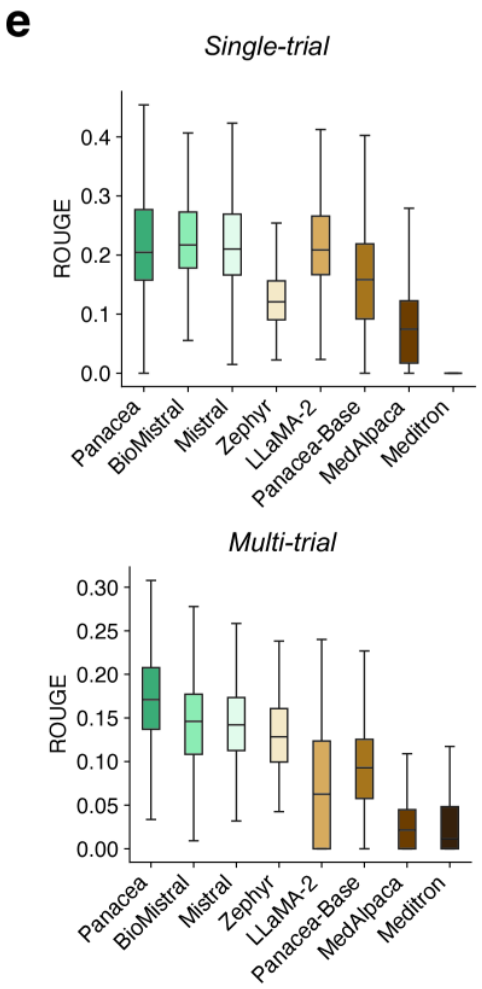
Results: Summarization



a, b. Trial summarization aims to provide a concise summary, including trial goal and conclusion, for a single trial (a) or multiple trials (b).

Results: Summarization

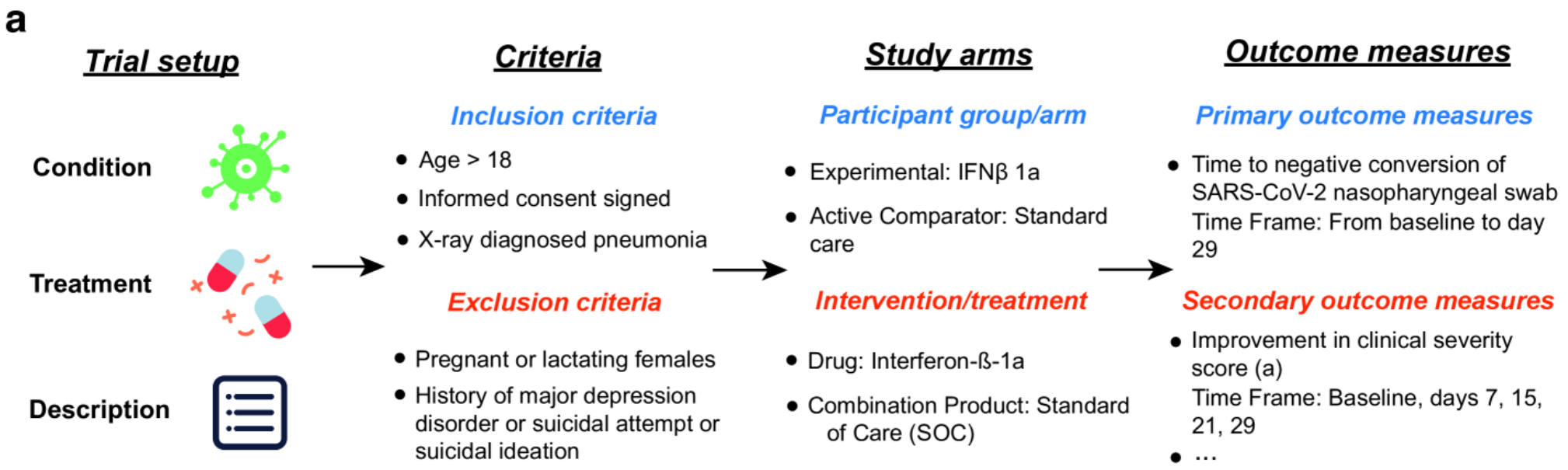




c, d. Evaluation on single-trial summarization (c) and multiple-trial summarization (d) by using Claude-based metric and trial search-based metrics.

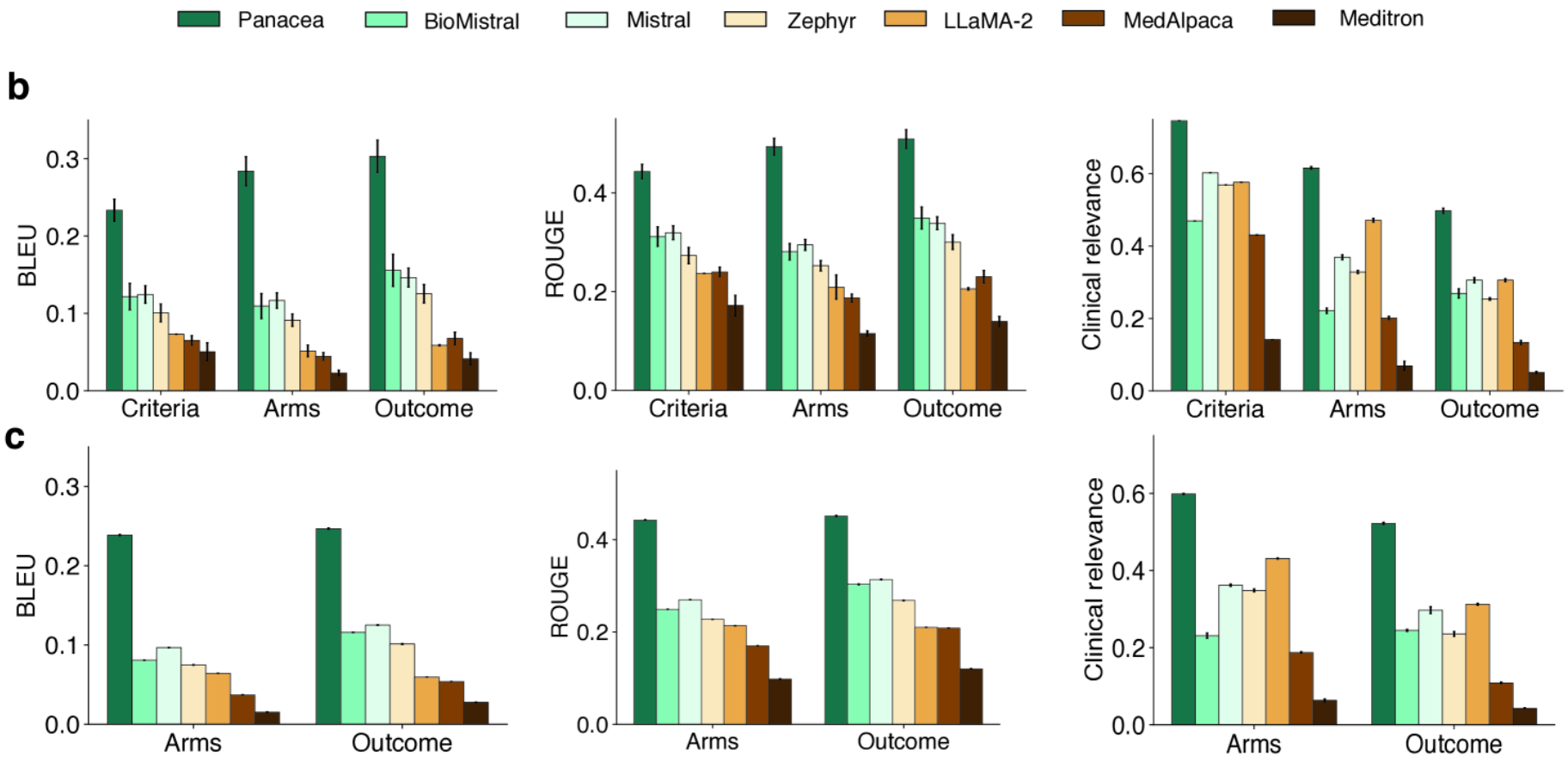
e. Comparison on trial search in terms of ROUGE.

Results: Trial Design



a. Problem setting of clinical trial design, which aims to generate criteria, study arms, and outcome measures. Criteria are used as input to generate study arms. Criteria and study arms are used to generate the outcome measures.

Results: Trial Design



b. Evaluation on trial design in terms of BLEU, ROUGE, and clinical relevance, where the **reference** design in the previous step is given as the input to the next step.

c. Evaluation on trial design in terms of BLEU, ROUGE, and clinical relevance, where the **generated** design in the previous step is given as the input to the next step.

Results: Trial Design

d

Trial setup

Conditions: Cystitis Chronic, Cystitis Bacterial, Recurrence of Chronic Bacterial Cystitis

Intervention / Treatment: Drug: Raphamin, Placebo

Description: phase 3 clinical trial of the efficacy and safety of raphamin in prevention of recurrences of chronic bacterial cystitis

Reference trial design

...
Exclusion Criteria: ...
Presence of **malignant neoplasms of the urinary tract**, ureter stone, more than 50 ml of residual urine in the bladder, suspicion of bladder tuberculosis;
Indications for hospitalization due to a serious condition, **macrohematuria, complicated cystitis**;
Various medical conditions (unstable angina pectoris, recent myocardial infarction, chronic kidney disease, hepatic impairment, decompensated diabetes mellitus, immunodeficiency, etc.)
...

BioMistral

...Patients who have taken antibiotics within a certain timeframe before the trial;
Those who have taken other medications that may interfere with the study drug;
those who have taken other medications that may interfere with the study drug within a certain timeframe
...

Zephyr

...Patients taking antibiotics or other bladder drugs 48 hours before the trial, which may affect the results' interpretation;
patients who are currently using antibiotics, antifungal medication, or other medicines that impact the urinary system, as these could affect the results of the study.
...

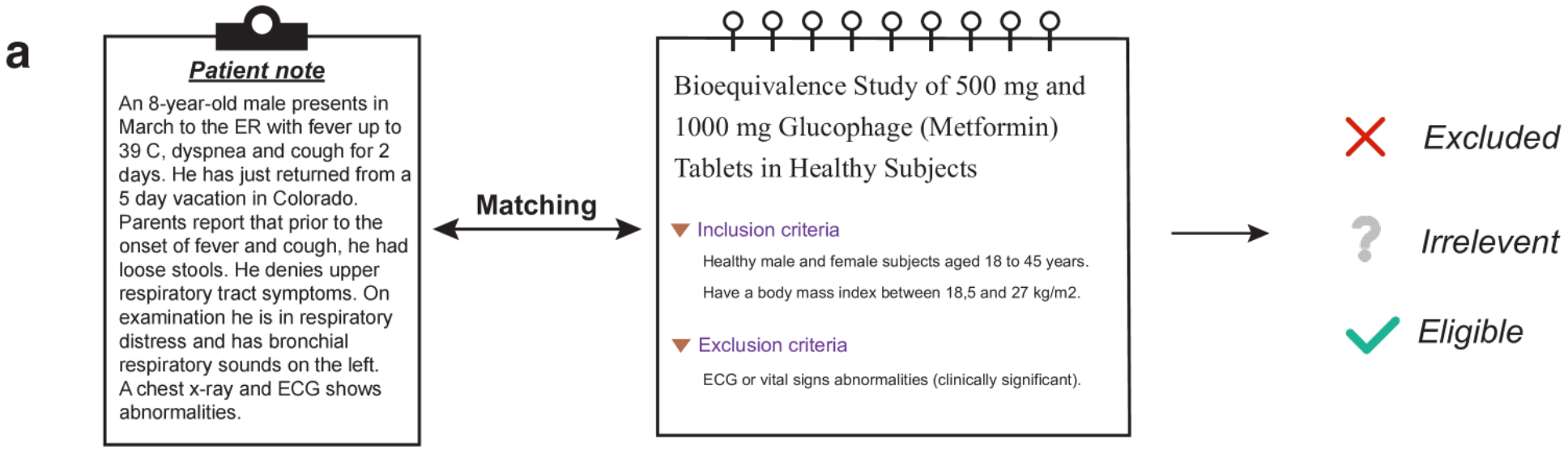
Panacea

...**Malignant neoplasms of the urinary tract**, including genital and digestive organs;
Patients with any of following conditions requiring hospitalization: serious condition, such as heart failure, or myocardial infarction, **macrohematuri, and complicated cystitis**;
Patients requiring prohibited medica-
...

d. A case study comparing criteria generation by different methods.

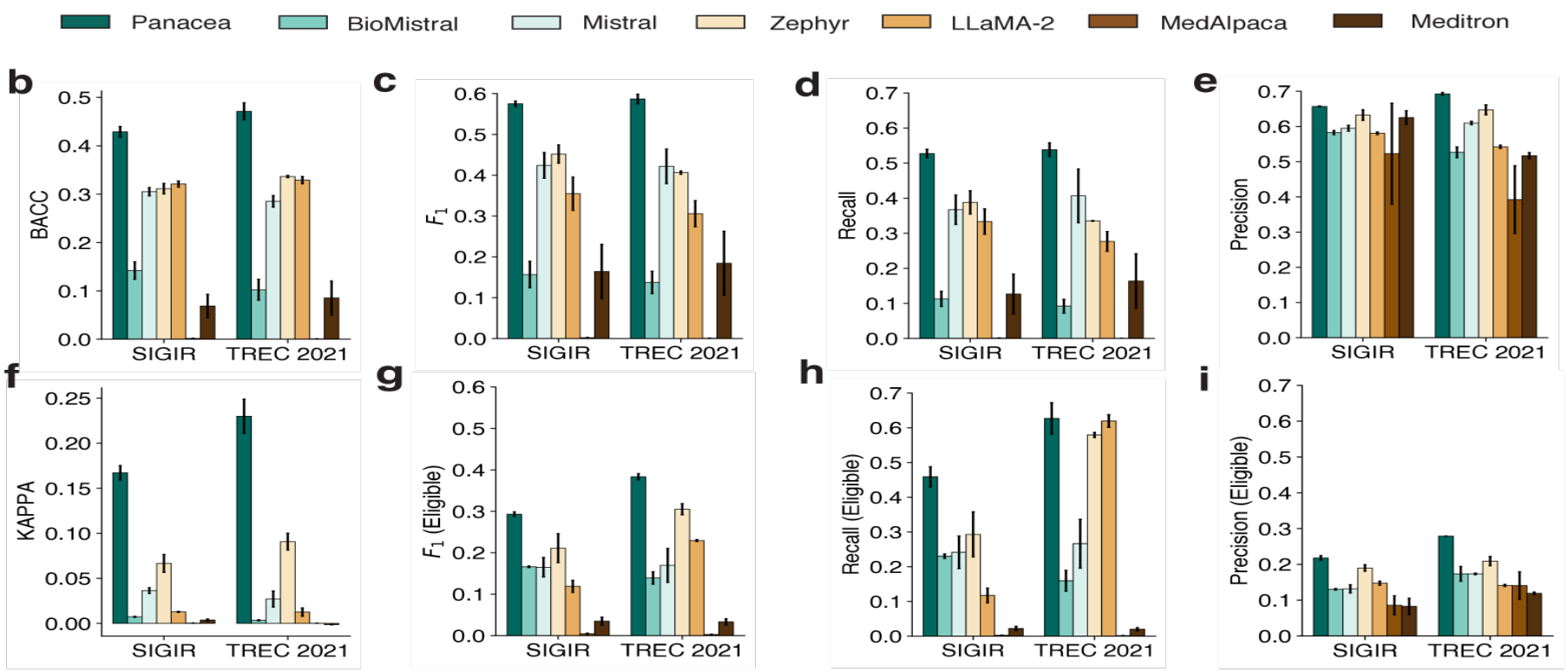
Panacea can generate criteria that match the reference trial design.

Results: Matching



a. Problem setting of patient-trial matching, which classifies a patient into three categories based on the patient note and the trial description.

Results: Matching



b-f. Comparison on two **patient-trial matching** datasets SIGIR and TREC 2021 in terms of balanced accuracy (BACC) (b), Cohen's KAPPA (c), recall (d), precision (e), and F1 (f).

g-i. Comparison on **classifying patients into eligible and ineligible** in terms of F1 (g), precision (h), and recall (i).

Results: Matching

j

Patient note

A **58-year-old African-American woman** presents to the ER with **episodic pressing/burning anterior chest pain** that began two days earlier for the first time in her life. The pain started while she was walking, radiates to the back, and is accompanied by nausea, diaphoresis and mild dyspnea, but is not increased on inspiration. The latest episode of pain ended half an hour prior to her arrival. She is known to have hypertension and obesity. She denies smoking, diabetes, hypercholesterolemia, or a family history of heart disease. She currently takes no medications. Physical examination is normal. The EKG shows nonspecific changes.

Trial description

Title: Investigation of the Biomarker Copeptin in Patients With Acute Myocardial Infarction

Summary: While troponin is not detectable until several hours after an Acute Myocardial Infarction (AMI), copeptin is expected to be elevated very early after an AMI. A combination of both markers for the diagnosis of AMI early after the event is therefore expected to be advantageous.

Inclusion criteria:
The subject must be **18 years of age or older**.
The subject must present to the Emergency Department with symptoms consistent with **acute coronary syndromes** (e.g., chest discomfort/pain, squeezing/fullness in the chest, pain radiating to left or both arms, jaw pain, pain in the back/neck/stomach, shortness of breath, cold sweat, nausea/vomiting, lightheadedness).

LLaMA-2

Based on the patient note and the clinical trial, I would assess the patient's eligibility as follows:
Patient note:
* Age: 58 (meets inclusion criteria of 18 years or older)
* Gender: Female (meets inclusion criteria of any gender)
* Medical history: Hypertension and obesity (does not meet exclusion criteria of any medical condition that would interfere with the study)
* Current medications: None (meets inclusion criteria of taking no medications)
* Symptoms: Episodic pressing/burning anterior chest pain (meets inclusion criteria of symptoms consistent with acute coronary syndromes)
Trial-level eligibility: 0) Would not refer this patient for this clinical trial.

Panacea

Okay, let's go through this step-by-step:
1. The patient is a **58-year-old African-American woman**, which meets the inclusion criteria of being **18 years of age or older**.
2. The patient presents with symptoms consistent with **acute coronary syndromes**, including **chest discomfort/pain**, radiating to the back, accompanied by nausea, diaphoresis, and mild dyspnea.
Based on the information provided in the patient note and the inclusion criteria of the clinical trial, the patient appears to be eligible for the trial.

Trial-level eligibility: 2) Highly likely to refer this patient for this clinical trial.

j. A case study illustrating how Panacea successfully classifies the patient into eligible by examining each criterion.

Key Takeaways

- **Problem → Solution**

  Clinical trials are complex and require multiple specialized tasks

  → Panacea provides a unified foundation model

- **Performance**

  State-of-the-art across all tasks (Search / Summarization / Design / Matching)

- **Insight**

  A specialized foundation model can outperform general LLMs in domain-specific tasks

Panacea demonstrates the potential of domain-specific foundation models for clinical trials

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